

Worksheet 102 · Spot on facts, pp. 226–227

Technologies that change society

The key facts at a glance — the connections are yours to make. Your basis for the Kolloquium.

1 Developments in the 21st century

Genetic engineering

CRISPR edits the genetic code precisely.

+ drought-resistant crops, new treatments

– morality; interference with DNA

Eco-technology

Solar, hydrogen and wind power are cheaper; more e-cars.

+ less fossil fuel, ocean clean-ups

– raw materials (lithium) harm the environment

AI & digital content

Prompts to large language models create texts, media, programs.

+ information re-used in myriad contexts

– where does content come from — can we trust it?

Connect: All three bring a hope and a new problem. Which development shows this most clearly — and why?

E.g. eco-technology: e-cars reduce the use of fossil fuels, but the lithium for their batteries creates new environmental problems — one solution produces the next problem.

2 Artificial intelligence (AI)

AI = the simulation of human intelligence in machines: learning, problem-solving, reasoning, perceiving.

Weak / narrow AI

single tasks, e.g. personal assistants, video games

Strong / general AI

complex, human-like, e.g. (book) self-driving cars, robots*

Impact

+ everyday life, education, health

– jobs, deepfakes, misinformation, copyright

*Most experts today call self-driving cars narrow AI; general AI means a still hypothetical human-level intelligence.

Connect: Why does checking data matter more than ever now that AI creates content?

AI can produce convincing texts and deepfakes, and it is often impossible to tell where content comes from. Checking sources is the only protection against manipulation and misinformation.

3 Augmented reality (AR) and virtual reality (VR) — both extended reality (XR)

AR

What it does adds digital overlays to the real world

Devices phone, tablet (camera)

Used in entertainment, medicine, shopping

VR

replaces the real world with a 360° environment

headset, motion controllers

gaming, tourism, education, medicine, warfare

Connect: How are AR, VR and AI related? Name one example you know (e.g. Virtual Helsinki, p. 230).

AR and VR are both types of XR; AI can make them smarter and more realistic (e.g. AI avatars, generated worlds). Example: Virtual Helsinki is VR, a digital twin of the city.

4 Utopia or dystopia — where will technology take us?

Utopia ou-topos = “no place”, eu-topos = “good place” (Thomas More, 1516).

An ideal society that criticises the present by showing a better world.

e.g. Butler, Erewhon (1872); Bellamy, Looking Backward (1888)

Dystopia dys-topos = “bad place”.

A worst-case society, often in the near future, that warns about contemporary problems.

e.g. Brave New World (1932), Nineteen Eighty-Four (1949), The Hunger Games (2008)

Connect: Which technology from boxes 1–3 could be the starting point of a dystopia? Describe that society in one or two sentences.

E.g. AI + VR: people spend most of their lives in virtual worlds whose content is selected by AI. Reality loses importance and opinions can be steered without anyone noticing — a society like Brave New World, kept happy and passive by technology.

Kolloquium Carousel

In Part 2 of the Kolloquium you answer questions on topics from other semesters — no notes. Train it now.

How it works

- 1 Double circle: inside candidates, outside examiners.
- 2 Examiner reads a card — 15 s to think, 2 min to answer.
- 3 Follow-up question (1 min), then rubric + two stars and a wish.
- 4 Outer circle moves on. Swap roles after two rounds.

Answer like a candidate

Define → Explain → Example → Evaluate

Question cards (follow-up in italics)

1 AR vs VR

Explain the difference between AR and VR.

Follow-up: Which will change everyday life more?

2 AI

What is AI? Weak vs strong AI.

Follow-up: Where does AI help — where does it worry you?

3 Trust

Why is it harder to know where online content comes from?

Follow-up: How can you protect yourself?

4 Genes

What does CRISPR make possible?

Follow-up: Where would you draw the line?

5 Terms

Where do utopia and dystopia come from?

Follow-up: Why are dystopias so popular today?

6 Big picture

Which technology is the most life-changing?

Follow-up: How will it affect your children's lives?

Examiner's sheet 0 = not yet · 1 = partly · 2 = mostly · 3 = fully

	Round 1	Round 2
Candidate		
Content — accurate facts and terms	/3	/3
Structure — clear line of thought	/3	/3
Language — topic words, range, accuracy	/3	/3
Interaction — reacts to the follow-up	/3	/3
Total	/12	/12

After the carousel — your Kolloquium note Which card was hardest? What would you need to answer it well next time?

Individual answers, e.g. card 4 (Genes): I need the exact definition of CRISPR, one concrete example (drought-resistant crops) and a clear position on where to draw the line (curing diseases yes, "designer babies" no).